Reliability of lookup table methods for clothing insulation: experimental verification using typical garments

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Abstract: Accurate quantification of clothing insulation is crucial for assessing human thermal balance and comfort. However, current research often relies heavily on standardized lookup tables, which may introduce significant biases. This study systematically investigates the discrepancies between estimated and measured clothing insulation values through a dual-method approach combining expert surveys and thermal manikin experiments. A total of 34 garments (20 upper-body and 14 lower-body) were configured into 16 representative ensembles. Clothing insulation values were estimated by 120 thermal comfort researchers based on standard references, and measured precisely using the "NEWTON" thermal manikin in a climate chamber (20-21 °C, 50% RH). The study found significant differences in insulation estimates among researchers when using standard references, with errors increasing as insulation values rose. These errors reached critical maxima of 69% for single garments and 33% for ensembles. The widely adopted McCullough equation was found to systematically underestimate clothing insulation for Chinese adult ensembles by 0-35%. Further analysis based on thermal comfort models demonstrated that such discrepancies in insulation inputs could lead to significant prediction deviations: up to 0.59 units on the PMV scale, 5 ℃ on the SET scale, and an average skin temperature error of approximately 1.74 ℃. These deviations may affect HVAC control strategies and occupant comfort assessments. This study contributes a refined insulation dataset for clothing ensembles and underscores the need for improved estimation methods. It highlights the importance of accurate insulation input for reliable thermal comfort predictions and provides empirical evidence to support future revisions of clothing insulation standards.

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Keywords: Clothing insulation, Thermal comfort, Estimated value, Measured value, Error analysis

Nomenclature

PMV	Predicted mean vote	clo	Clothing insulation (clo)
SET	Standard effective temperature (℃)	I_t	Total clothing insulation (clo)
UCB	University of California, Berkeley	I_{cl}	Intrinsic clothing insulation (clo)
HVAC	Heating, Ventilation, and Air Conditioning	I_{clu}	Effective clothing insulation (clo)
ISO	International Organization for Standardization	f_i	Ratio of the surface area of segment *i* to the total surface area of the manikin
ASHRAE	American Society of Heating, Refrigerating and Air-Conditioning Engineers	$T_{sk,i}$	The manikin's skin surface temperature of segment *i* (℃)
GB/T	GuoBiao/Tui, Chinese Standard	T_a	Ambient air temperature (℃)
ASTM	American Society for Testing and Materials	H_i	The local heat loss from segment *i* (W/m^2)
TSV	Thermal sensation vote	I_a	Thermal insulation of the nude manikin (clo)
T_{sk}	Skin temperature (℃)		

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1. Introduction

Clothing insulation stands as a pivotal factor in human thermal comfort, fundamentally influencing heat exchange between the human body and its environment [1-3]. The scientific quantification of this parameter traces back to Gagge et al. [4], who established the "clo (clothing thermal insulation)" unit (1 clo = 0.155 $m^2 \cdot K/W$) as a standardized measure of thermal resistance. This seminal work laid the foundation for systematic clothing insulation assessment. Building upon this, Fanger's established the Predicted Mean Vote (PMV) model [5] in the 1970s incorporated clothing insulation as one of six core parameters in human heat balance calculations, cementing its role in thermal comfort evaluation. Additionally, clothing insulation is a critical input variable in several other human thermal comfort evaluation indices and prediction models, such as the Standard Effective Temperature (SET) model [6], Stolwijk model [7], Fiala model [8], Tanabe model [9], and University of California, Berkeley (UCB) model [10], all of which confirm the importance of clothing insulation in assessing and predicting human thermal comfort.

Fig. 1 illustrates the impact of changes in clothing insulation on the predictive results of the PMV model at a metabolic rate level of 1.1 met [11], with PMV calculations performed using the computation program provided by UCB [12]. Specifically, under summer conditions, when the clothing insulation value increases from 0.3 clo to 0.6 clo, it can cause a shift in the neutral temperature of more than 1.5 °C or result in a 0.7 units scale difference in PMV values. Therefore, the accurate input of clothing insulation values for individuals in thermal comfort research is a prerequisite to ensure the accuracy of model prediction results, which plays a crucial role in the evaluation of the indoor thermal environment and the design and operation of Heating, Ventilation, and Air Conditioning (HVAC) systems, and thus directly affects the energy consumption of buildings [13,14].

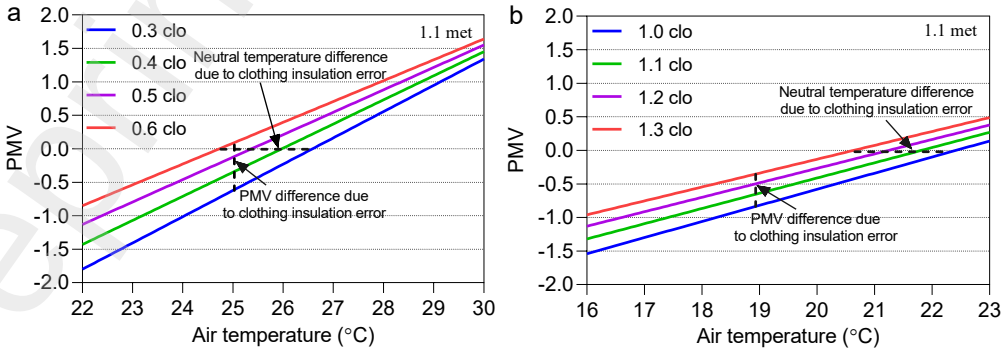


Fig. 1. Effects of clothing insulation change on PMV error (a) summer; (b) winter

However, clothing insulation values cannot be directly measured like environmental

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parameters through physical instruments; accurate insulation values require testing with a thermal manikin [15-18], a process that demands high requirements for conditions and environment, making its feasibility not strong [19]. Therefore, researchers typically refer to standard tables to determine the insulation values of typical clothing [20,21], a method widely used in the field of thermal comfort due to its convenience [22-24]. In existing thermal comfort standards, ISO (International Organization for Standardization) 9920-2007 [16], ISO 7730-2005 [20], ASHRAE (American Society of Heating, Refrigerating and Air-Conditioning Engineers) 55-2020 [21], and GB/T (Chinese Standard) 50785-2012 [25] (which references ISO 7730-2005 values) are commonly referenced for clothing insulation values, with each standard providing clear regulations on the calculation of clothing insulation. There are two main methods for obtaining the insulation value of an ensemble: (1) matching the typical suit insulation provided in the corresponding standards for the surveyed clothing; (2) simply stacking the insulation of single garments and calculating the ensemble insulation using the linear regression model provided by McCullough et al. [26]. The clothing insulation data in existing thermal comfort databases, such as the SCATs database [27], ASHRAE global thermal comfort database [23,28], and Chinese thermal comfort database [14], are obtained using the above methods. Jia et al. [29,30] used a questionnaire survey and lookup table method to get different clothing insulation values for passengers waiting in airports. Zhao et al. [31] adopted a similar approach to investigate the insulation values of customers in cold-region shopping malls in China across different seasons, obtaining the relationship between temperature and clothing insulation values. Based on this method, Su et al. [32] explored the clothing behavior of residents in different climatic zones in China and established a comprehensive model for clothing insulation, providing valuable references for achieving a balance between thermal comfort and energy consumption.

The clothing insulation databases provided in existing standards are based on experimental results from decades ago [26,33,34]. With the advancement of material engineering and the increasing diversity of clothing materials, there is an urgent need to update and expand the existing clothing insulation databases. Numerous studies have supplemented and updated the insulation values for different populations in recent years to meet the demand for diverse clothing insulation. In 2015, Havenith et al. [35] tested 52 types of clothing from 7 countries, providing additional data for non-Western populations in the ASHRAE 55-2020 [21] standard. In 2021, Smallcombe et al. [36] tested 69 sets of clothing for Western populations, updating and refining the insulating performance data in standards such as ISO 9920-2007 [16], ISO 7730-2005 [20], and ASHRAE 55-2020 [21]. In China, Tang et al. [37-39] based on the winter dressing characteristics of the elderly population, used a thermal manikin to test and supplement

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the insulation values of 34 typical sets of clothing, establishing a database for elderly clothing insulation and updating the typical clothing insulation database in the GB/T 50785-2012 [25]. Despite these studies updating existing standards, there are still differences in the types of clothing insulation databases and reference values provided by different standards, and the calculation methods for suit insulation also vary [40]. This raises two key questions: How much difference exists between the insulation values obtained from different standards and the actual measured values? Can the calculation methods for suit insulation in existing standards effectively apply to the prediction of current clothing insulation values in China? These questions have not yet been clearly answered in existing research and require further exploration and verification.

It is worth noting that the clothing insulation values obtained through the lookup method are generally estimated and may have limited accuracy. Lee et al. [41] experimentally measured the insulation of suits composed of shirts, pants, and shoes with different materials, and found that while the standard estimated the suit insulation value to be 0.61 clo, the actual measured values ranged from 0.65 to 0.83 clo, indicating a discrepancy between estimated and measured values. Further research by Havenith et al. [42] discovered that the clothing insulation values obtained through the lookup method were not sufficient to ensure the prediction accuracy of the PMV model within a 0.3 scale unit. Tang et al. [37] study also pointed out that the estimated clothing insulation values in the ISO 7730-2005 [20] were inconsistent with the actual test results of older people, with estimation errors potentially causing a predicted average skin temperature deviation of about 1.3 °C or a heat loss deviation of about 26%. These research results confirm the differences between estimated and measured values, and such differences can affect the prediction results of thermal comfort models and may introduce bias. Currently, there is a relative lack of quantitative research on the impact of errors between estimated and measured clothing insulation values on model prediction accuracy. Moreover, in real scenarios, there is a variety of clothing types, including office wear, casual wear, and sportswear, and it is not yet clear whether the differences between estimated and measured values are consistent across different types of clothing. Therefore, further research on the differences between estimated and measured insulation values for different clothing types is of great significance for improving the prediction accuracy of thermal comfort models.

This study aims to delve into the differences between estimated and measured values of clothing insulation and the impact of these differences on the prediction accuracy of thermal comfort models. We selected 16 representative sets of clothing (including 20 upper garments and 14 lower garments), covering three different types of attire: office

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ensemble, casual ensemble, and sports ensemble. The study is divided into the following steps: First, we distributed questionnaires to scholars who have long been engaged in thermal comfort research, asking them to estimate the insulation values of single and suit items listed in the questionnaire according to standards, with the aim of collecting estimated data from different scholars based on existing standards; Second, we used a thermal manikin to precisely measure the insulation of each set of clothing to obtain measured values, ensuring that we could conduct an objective quantitative analysis of clothing insulation; Finally, we compared the estimated values with the measured values, analyzed the differences between the two, and assessed the specific impact of these differences on the prediction accuracy of thermal comfort models.

Through this study, we will answer the following three core questions: i) What are the differences between the estimated and measured insulation values for different types of clothing? ii) How reliable are the current standards for calculating the insulation of suit clothing? iii) What is the specific impact of the differences between estimated and measured clothing insulation values on the prediction accuracy of thermal comfort prediction models?

2. Methods

The overall procedure of this study is shown in Fig. 2. First, a total of 16 typical ensembles, which contained 34 individual garments, were selected. Then, make a questionnaire with the photos of the selected clothing, as shown in Fig. 2. Questionnaires were sent by email to a number of scholars with more than 10 years of research experience in the field of thermal comfort, and the researchers assigned values to individual garments and ensemble according to the corresponding criteria. After strict screening, 120 valid questionnaires were finally collected, which provided a reliable data basis for subsequent analysis. Secondly, the clothing insulation of reported garments and ensembles was tested using manikin experiments to widen the database of clothing insulation. Additionally， the estimated and measured clothing insulation are compared with the results of PMV [5] and SET [6] models.

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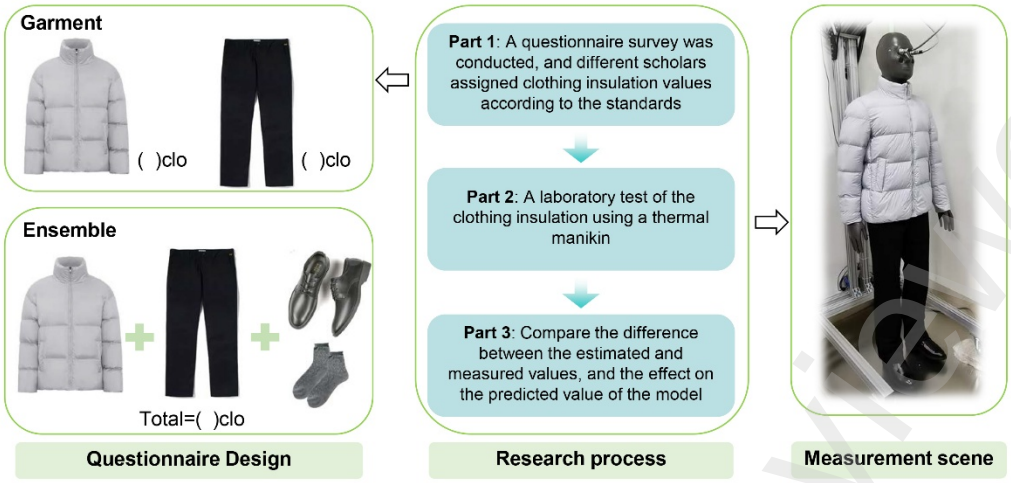


Fig. 2. Flowchart of this study

2.1. Choice of typical garments and clothing ensembles

This study focuses on college students as the research subjects and records common types of clothing through photography from April to December 2022. Ultimately, 34 typical garments were selected for testing, of which 20 were installed on the top, and 14 were installed on the bottom. Fig. 3 shows photos of selected research and tested upper garment types, including short sleeves, long sleeves, jackets, sweaters, down jackets, etc. Fig. 4 shows pictures of lower garment types that were researched and tested, including shorts, sports trousers, jeans, casual trousers, etc. Each of the ensembles is described in detail in Appendix A. All tested garments are sized to fit the thermal manikin. From the tested single pieces of clothing, 16 different combinations were formed for suit testing, as shown in Fig. 5. These include three types: sports ensemble (four sets), office ensemble (four sets), and casual ensemble (eight sets). Detailed information on the ensembles can be found in Appendix B.

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Fig. 3. Clothing insulation measurement photos of the upper clothes (U1: T-shirt 1; U2: T-shirt 2; U3: Polo shirt; U4: Short-sleeved shirt; U5: Sports long sleeve 1; U6: Sports long sleeve 2; U7: Long-sleeved underwear; U8: Long-sleeved shirt 1; U9: Long-sleeved shirt 2; U10: Sweatshirt; U11: Hoodies; U12: Sports jacket; U13: Denim jacket; U14: Suit jacket 1; U15: Suit jacket 2; U16: Sweater; U17: Outdoor jacket; U18: Down jacket 1; U19: Down jacket 2; U20: Down jacket 3)



Fig. 4. Clothing insulation measurement photos of the lower clothes (L1: Shorts; L2: Light skirt; L3: Long legs underwear; L4: Sweatpants + shorts; L5: Sweatpants (thin); L6: Sweatpants (middle); L7: Sweatpants (thick); L8: Coveralls; L9: Suit pants; L10: Jeans (thin); L11: Jeans (thick); L12: Casual pants (thin); L13: Casual pants (middle); L14: Casual pants (thick))

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Fig. 5. Clothing insulation measurement photos of ensembles by using the thermal manikin Newton

2.2. Climate chamber and environment measurements

The thermal insulation values of the clothing were measured using an electrically heated manikin housed in a climate chamber. The environmental conditions within the climate chamber were strictly controlled according to the requirements specified in ISO 15831-2004 [15]. Specifically, the air temperature in the chamber was regulated between 20.0-21.0 ℃, the relative humidity was approximately 50%, and the air velocity was less than 0.1 m/s. These parameters were continuously monitored using HD 32.2 TCA (air temperature and globe temperature accuracy ± 0.1 °C, relative humidity accuracy ± 2.5%, and air velocity accuracy ± 0.05 m/s) during all manikin tests to ensure compliance with the standard. The sensor was placed 0.5 m from the manikin. Ambient air temperature was continuously monitored using three sensors positioned adjacent to the ankles, the mid trunk, and the head (vertical heights of 0.1 m, 0.6m, 1.1m, and 1.7m). Airspeed was less than 0.1 m/s in the chamber. The control accuracies for the air temperature and relative humidity were ±0.2 ℃ and ±5%, respectively. The mean radiant temperature was close to the air temperature due to the heavily insulated chamber envelope and inner surfaces made of stainless steel. The size of the climate chamber is 4.5 m (length) × 3.9 m (width) × 2.7 m (height), which meets the requirements for test space dimensions in ISO 15831-2004 [15], as shown in Fig. 6.

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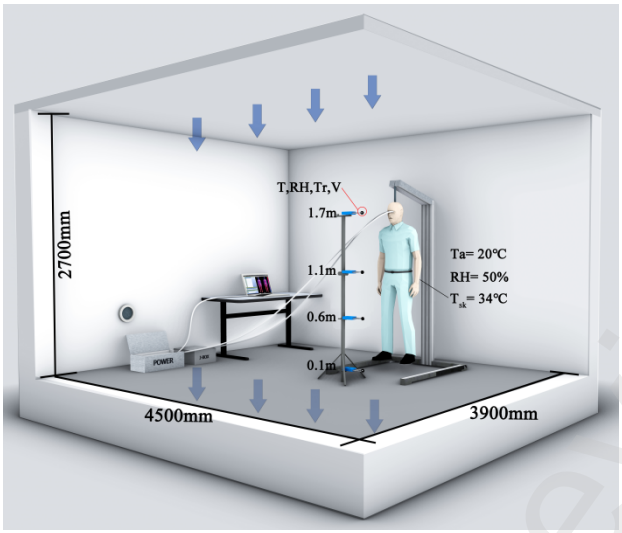


Fig. 6. Climate chambers test scenes

2.3. Thermal manikin

Existing studies and related standards have shown that accurate clothing insulation values can be obtained by using a thermal manikin [17,43]. Therefore, the measurement method of the thermal manikin was adopted in this study. The thermal manikin used at Xi'an University of Architecture and Technology. The test scenario is shown in Fig. 6. The height of the thermal manikin is 1.68 m, with a body surface area of 1.68 m^2. The manikin consists of 26 body segments, each independently controlled regarding surface temperature and monitored for heat flow. The manikin's body proportions correspond to those required for standard sizes of garments. According to ISO 15831-2004 [15], the Constant Temperature Control Mode of the manikin and the surface temperature of the manikin zones were controlled at 34.0 ℃. Its accuracy was within ±2% of the value for the average power supplied to each body segment of the manikin during the test period. The manikin was calibrated according to ASTM (American Society for Testing and Materials) Standard F1291-2016 [17] and ISO 15831-2004 [15] prior to experiments, ensuring <2% measurement deviation.

2.4. Test procedure

The test procedure is shown in Fig. 7. The test procedure is based on standard ISO 15831-2004 [15] and ISO 9920-2009 [16]. Prior to testing, the garments were conditioned either at 20 ℃ and 50% RH in the climate chamber for at least 12 h. All clothing ensembles were tested in the standing posture. The manikin is dressed in the clothing ensemble to be tested, with each garment arranged on the appropriate part of its body as in practical use. Each test lasted 70 min for a single garment and 90 min for clothing ensembles, and ended after both the heat flow and the skin temperature were stable. A minimum of 30 minutes of stability was required. Each ensemble was repeated

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three additional times to ensure accuracy and compare the results before the first and second calculations, and the error was less than 4% by the end of the test; otherwise, repeat the third test. The temperature and heat losses were recorded at 10-second intervals. During the clothing insulation test, the clothing is normally placed on the thermal manikin to keep the clothing flat and reduce wrinkles.

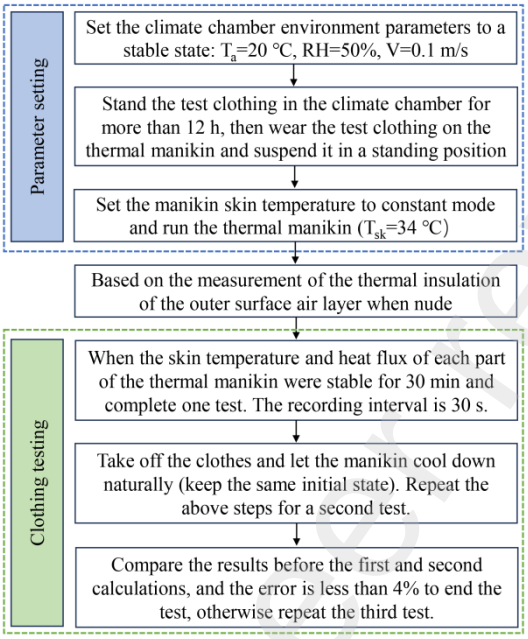


Fig. 7. Clothing insulation test procedure

2.5. Calculation of clothing insulation values

Due to the varying impact of clothing on different areas of the thermal manikin, it is necessary to apply weighted processing to varying regions during calculations. The common calculation methods include the parallel method and the series method. Research has found that the series method can introduce significant bias when the distribution of clothing insulation is uneven [44]. Therefore, this study opts for the parallel method to calculate the clothing insulation values. Clothing insulation can be divided into total insulation (I_t), intrinsic insulation (I_{cl}), and effective insulation (I_{clu}), as shown in Fig. 8. Data from the last 30 minutes of the stable state were used for insulation calculation. The clothing insulation reference values provided in the standards ISO 7730-2005 [20] and ASHRAE 55-2020 [21] offer effective insulation values for garments, while the insulation of ensembles was expressed by intrinsic insulation. The values presented in this study also follow the standard format.

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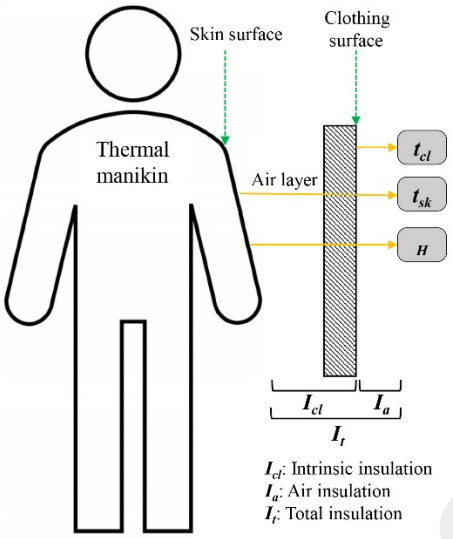


Fig. 8. Different expressions of clothing insulation

Calculations for total insulation (I_t). Total insulation values are measured directly with a manikin. Total insulation is the thermal resistance from the skin surface to the environment, including all clothing, enclosed air layers, and boundary air layers. Total insulation values were calculated according to the parallel method and were calculated by Eq. 1.

$$I_t = \frac{((\sum f_i \cdot T_{sk,i}) - T_a)}{0.155 \times \sum f_i \cdot H_i} \quad (1)$$

Where I_t represents the total thermal insulation of clothing plus air layer, clo (1clo=0.155 m^2 K/W); f_i represents the ratio of the surface area of segment i to the total surface area of the manikin; $T_{sk,i}$ is the manikin's skin surface temperature of segment i, ℃; T_a is the ambient air temperature, ℃; H_i represent the local heat loss from segment i, W/m^2.

Calculations for intrinsic insulation (I_{cl}). The insulation of the nude manikin was determined based on the value of the surface air insulation heat resistance. The intrinsic clothing insulation I_{cl} was determined by Eq. 2.

$$I_{cl} = I_t - \frac{I_a}{f_{cl}} \quad (2)$$

Where I_{cl} represents the intrinsic clothing insulation (clo), I_a represents the thermal insulation of the nude manikin (clo), and f_{cl} is the clothing area factor. The clothing area factor can be estimated according to the formula given in ISO 9920-2007 [16], as shown in Eq.3.

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$$f_{cl} = 1 + 0.28 \cdot I_{cl} \quad (3)$$

Calculations for air insulation (I_a). Air insulation is the thermal resistance of the air layer on the surface of the nude manikin. It can be calculated by Eq. 4.

$$I_a = \frac{T_{sk} - T_a}{0.155 \cdot H_{nude}} \quad (4)$$

Calculations for effective insulation (I_{clu}). Effective insulation is the increased thermal resistance provided by a single garment. It was determined by subtracting the resistance of the air insulation from the total insulation by Eq. 5.

$$I_{clu} = I_t - I_a \quad (5)$$

3. Results

3.1. Difference between the estimated clothing insulation of a garment and the measured values

In this study, we conducted precise measurements of the insulation values of all individual garments using a thermal manikin, with specific materials and test results detailed in Appendix A. Additionally, we collected 120 survey questionnaires from various researchers, which involved their estimations of clothing insulation based on reference standards (ISO 9920-2007 [16], ISO 7730-2005 [20] and ASHRAE 55-2020 [21]). Fig. 9 displays the distribution of estimated insulation values for single upper garments by different researchers compared to the actual measured effective insulation values, with the boxplot reflecting the distribution of these estimates and the red square representing the actual measured values. Our survey results revealed significant differences in the insulation values of single upper garments provided by different researchers based on reference standards, and these differences increased with higher insulation values. Table 1 presents the statistics for the estimated and measured insulation values of upper garments.

Specifically, the estimated insulation values for upper garments ranged from 0.05 to 0.80 clo according to different researchers' reference standards, while the actual measured values for single upper garments ranged from 0.11 to 0.88 clo. The average error rate between estimated and actual insulation values was 21%, with a maximum error rate reaching 69%. The standard deviation of the estimated insulation values for all upper garments ranged from 0.03 to 0.10 clo. Taking short sleeves as an example, the estimated range was 0.11 to 0.15 clo, while the actual measured values for different types of short sleeves ranged from 0.11 to 0.18 clo. In the case of suit jackets, the difference between the maximum and minimum values given by different researchers was as high as 0.52 clo, with the actual measured insulation values ranging from 0.31

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to 0.33 clo. The estimation differences for down jackets were also significant, with a standard reference value of 0.55 clo [20,21], but considering factors such as the filling amount, length, and material of the down jackets, the maximum difference in estimates for hooded long down jackets reached 0.40 clo, with a difference of 0.24 clo between the estimated average value and the actual measured value, and a significant difference between the estimated average and the actual measured values.

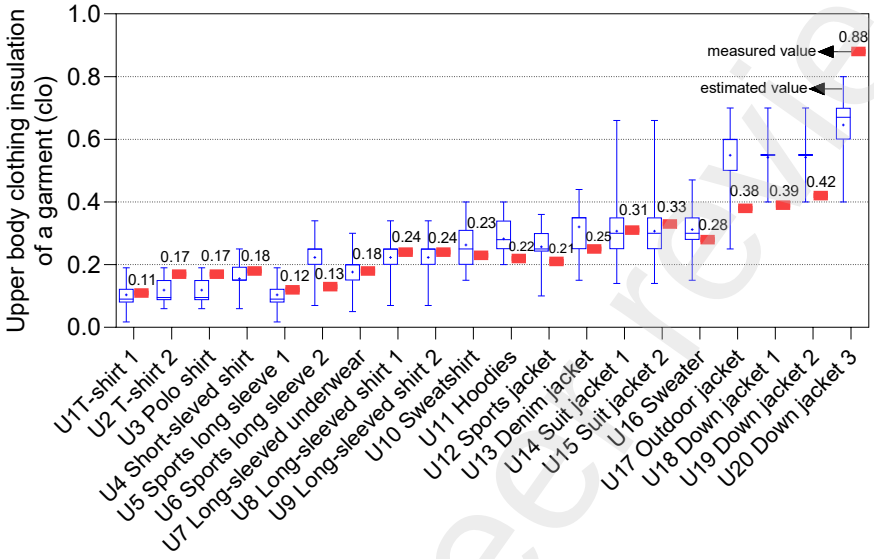


Fig.9. Comparison of upper body clothing insulation estimate and measured value

Table 1 Statistical table of upper body clothing insulation estimation and measured value

Number	Estimated value			Measured value (clo)	Error	Significance
	Minimum (clo)	Maximum (clo)	Average ± SD			
U1	0.05	0.19	0.11±0.03	0.11	0%	n.s.
U2	0.06	0.19	0.12±0.03	0.17	29%	***
U3	0.06	0.19	0.12±0.03	0.17	29%	***
U4	0.06	0.25	0.15±0.04	0.18	17%	***
U5	0.05	0.19	0.11±0.04	0.12	8%	*
U6	0.07	0.34	0.22±0.06	0.13	69%	***
U7	0.05	0.30	0.18±0.05	0.18	0%	n.s.
U8	0.07	0.34	0.22±0.06	0.24	8%	n.s.
U9	0.07	0.34	0.22±0.06	0.24	8%	n.s.
U10	0.15	0.40	0.26±0.06	0.23	13%	**
U11	0.20	0.40	0.28±0.05	0.22	27%	***
U12	0.10	0.36	0.25±0.05	0.21	19%	***
U13	0.15	0.44	0.32±0.06	0.25	28%	***
U14	0.14	0.66	0.31±0.08	0.31	0%	n.s.
U15	0.14	0.66	0.31±0.08	0.33	6%	n.s.
U16	0.15	0.47	0.31±0.05	0.28	11%	***

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U17	0.25	0.70	0.55±0.10	0.38	45%	***
U18	0.40	0.70	0.54±0.04	0.39	38%	***
U19	0.40	0.70	0.54±0.04	0.42	29%	***
U20	0.40	0.80	0.64±0.07	0.88	27%	***

(SD=standard deviation; Error=|measure value - estimated average value|/measured value; ***p<0.0001, **p<0.001, *p<0.01, n.s.=no significance difference)

Fig. 10 and Table 2 together present the distribution and statistical situation of the estimated and measured effective insulation values of single-piece lower garments by 120 researchers. The test results indicate that the estimated insulation range for lower garments, as given by different researchers based on reference standards, is between 0.02 and 0.35 clo, while the actual measured insulation values range from 0.07 to 0.30 clo. Among all lower garments, the average error rate between estimated and measured values is 15%, with a maximum error rate reaching 30%. The estimated standard deviation of insulation is between 0.01 and 0.06 clo, and the maximum error between estimated and measured values is 0.08 clo. Although this error rate is significant, it is much lower than the error rate between the estimated and measured values of upper garments (which can reach up to 69%), and the overall difference in the clothing insulation of suit garments is largely influenced by the errors in the upper garments.

Further analysis reveals that the insulation estimation range for sweatpants (L5, L6, and L7) is from 0.08 to 0.40 clo. For the same pair of sports pants, the difference between the maximum and minimum values given by different researchers is as high as 0.28 clo. Actual measurement results show that the measured insulation values of sweatpants of different thicknesses range from 0.19 to 0.26 clo. The estimated insulation values of casual pants also show significant differences. After considering factors such as the thickness and material of casual pants, the maximum difference between the measured values and the estimated average values given by different researchers is 0.08 clo, with an error rate ranging from 10% to 30%. Statistical analysis finds significant differences between the estimated and measured values of casual pants.

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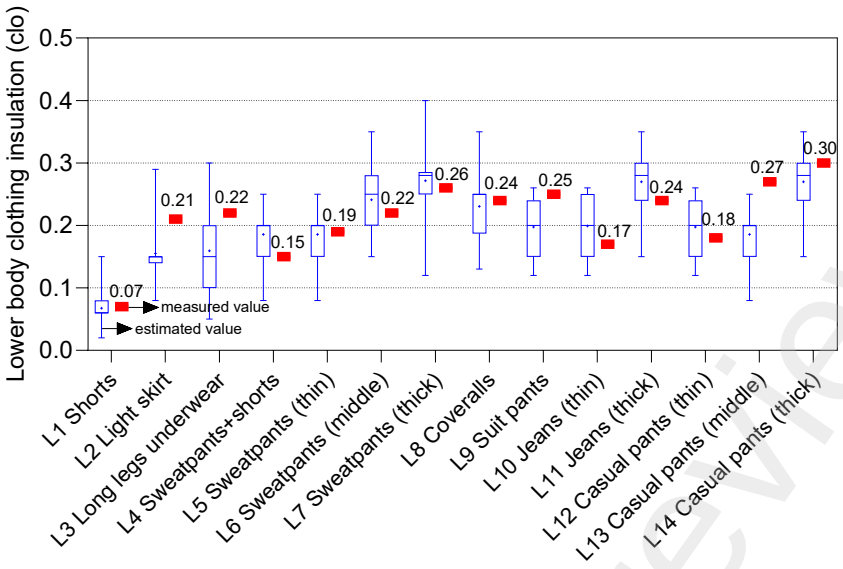


Fig. 10. Comparison of lower body clothing insulation estimate and measured value

Table 2 Statistical table of lower body clothing insulation estimation and measured value.

Number	Estimated value			Measured value (clo)	Error	Significance
	Minimum (clo)	Maximum (clo)	Average ± SD			
L1	0.02	0.15	0.06±0.01	0.07	14%	n.s.
L2	0.08	0.29	0.16±0.04	0.21	24%	***
L3	0.05	0.30	0.16±0.06	0.22	27%	***
L4	0.08	0.25	0.19±0.03	0.15	27%	***
L5	0.08	0.25	0.19±0.03	0.19	0%	n.s.
L6	0.15	0.35	0.24±0.04	0.22	9%	**
L7	0.12	0.40	0.27±0.04	0.26	4%	n.s.
L8	0.12	0.35	0.23±0.05	0.24	4%	n.s.
L9	0.12	0.26	0.20±0.03	0.25	20%	***
L10	0.12	0.26	0.20±0.05	0.17	18%	***
L11	0.15	0.35	0.27±0.05	0.24	13%	***
L12	0.12	0.26	0.20±0.04	0.18	11%	**
L13	0.08	0.25	0.19±0.04	0.27	30%	***
L14	0.15	0.35	0.27±0.05	0.30	10%	***

(SD=standard deviation; Error=|measure value - estimated average value|/measured value; ***p<0.0001, **p<0.001, *p<0.01, n.s.=no significance difference)

3.2. Difference between the estimated clothing insulation of the ensemble and the measured values

Fig. 11 presents the distribution of estimated versus measured basic insulation values for suit clothing by 120 scholars. The boxplot reflects the distribution of estimates for suit clothing insulation, with the square indicating the actual measurements from a thermal manikin. Survey questionnaire results show that when estimating the insulation

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of suit clothing, about 40% of scholars choose to directly sum the effective insulation values of individual garments to determine the basic insulation of the suit, while the remaining 60% apply McCullough et al. [45] correction equation after summing the insulation of individual garments, as shown in Eq. 6.

$$I_{cl} = 0.835 \sum I_{clu,i} + 0.161 \quad (6)$$

$\sum I_{\mathrm{clu},i}$ represents the sum of individual garment insulations in clo.

In this study, we measured the basic insulation values for three types of typical suit clothing: sports ensemble, office ensemble, and casual ensemble, with measured ranges of 0.35 to 1.1 clo, 0.56 to 1.33 clo, and 0.37 to 1.72 clo, respectively.

Table 3 provides a statistical analysis of the errors between the estimated and measured values of ensemble clothing insulation by different scholars. Overall, the estimated clothing insulation values for 5 sets (about 31%) were higher than the measured values, while for 9 sets (about 56%) they were lower. The standard deviation of the insulation estimates for all suits ranged from 0.05 to 0.19, with the maximum error between estimated and measured values reaching 0.21 clo. Significant differences existed between estimated and measured values for all suits, with an average error rate of 14% and the highest error rate of 33% observed in suit T10. The larger the insulation value of the suit clothing, the greater the standard deviation in estimates by different researchers. For example, in suits T4, T8, and T16, the standard deviation of estimates by different researchers was 0.14 clo, 0.11 clo, and 0.19 clo, respectively, but the error rates between their estimates and measured values were relatively small. This indicates that when the insulation value of the subject's suit clothing is larger, the error in the insulation value taken is also larger, which may lead to increased errors between model prediction accuracy and actual thermal sensation vote values at the neutral to slightly cool end [46-48].

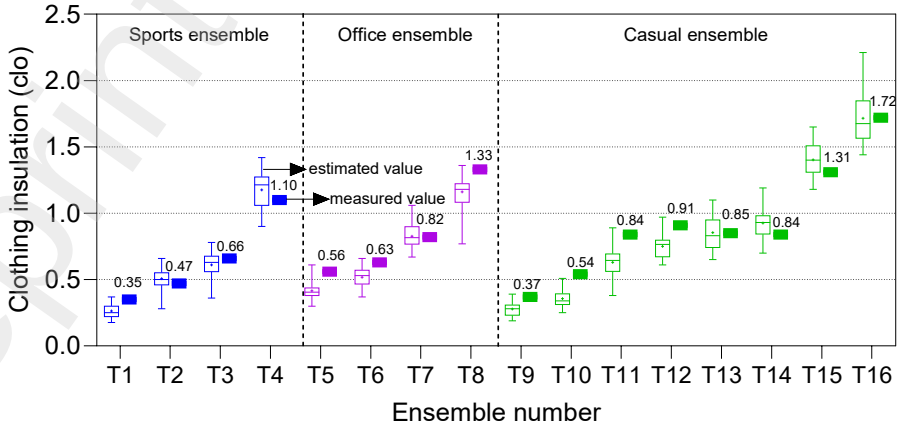


Fig.11. Comparison of ensemble clothing insulation estimate and measured value

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Table 3 Statistical table of thermal resistance estimation and the measured value of ensembles.

Number	Estimated value			Measured value (clo)	Error	Significance
	Minimum (clo)	Maximum (clo)	Average ± SD			
T1	0.18	0.37	0.26±0.05	0.35	26%	***
T2	0.28	0.66	0.51±0.08	0.47	9%	**
T3	0.36	0.78	0.61±0.09	0.66	8%	***
T4	0.90	1.42	1.18±0.14	1.10	7%	***
T5	0.30	0.61	0.42±0.07	0.56	25%	***
T6	0.37	0.66	0.52±0.07	0.63	17%	***
T7	0.67	1.06	0.83±0.10	0.82	1%	*
T8	0.77	1.36	1.16±0.11	1.33	13%	***
T9	0.19	0.39	0.28±0.05	0.37	24%	***
T10	0.25	0.51	0.36±0.06	0.54	33%	***
T11	0.38	0.89	0.63±0.10	0.84	25%	***
T12	0.61	0.97	0.75±0.09	0.91	18%	***
T13	0.65	1.10	0.85±0.13	0.85	0%	*
T14	0.70	1.19	0.93±0.12	0.84	11%	***
T15	1.18	1.65	1.40±0.12	1.31	7%	***
T16	1.44	2.21	1.72±0.19	1.72	0%	*

(SD=standard deviation; Error=|measure value - estimated average value|/measured value; ***p<0.0001, **p<0.001, *p<0.01)

4. Discussions

4.1. Comparison of clothing insulation of single garment with existing standards

In previous thermal comfort studies, the insulation values of individual garments were typically determined by referring to the ISO 7730-2005 [20] and ASHRAE 55-2020 [21] standards. This study compared the insulation test results of garments such as short sleeves, jackets, down coats, sports pants, and casual pants with the recommended values of ISO 7730-2005 and ASHRAE 55-2020, as shown in Fig. 12. The research found significant differences in the recommended values for the same types of clothing across different standards, with the ASHRAE 55-2020 standard generally providing lower values than the ISO 7730-2005 standard. For example, the reference values recommended by the standards are 0.09 clo (ISO 7730-2005) and 0.08 clo (ASHRAE 55-2020) for short sleeves, while the actual measurements show that the insulation values of short sleeves range from 0.11 to 0.77 clo, with significant differences in insulation values among different materials, and all test results exceed the standard recommended values. For jackets and down coats, the measured values are generally lower than the recommended values of both standards (except for long down coats), indicating that the standard recommended values may be overestimated, but there are still significant differences in insulation values between jackets of different materials

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and down coats with different filling amounts. The thickness and material differences between sports pants and casual pants also lead to significant differences between the measured insulation values of the clothing and the standard recommended values.

The main reasons for the differences between the actual measured insulation values and the standard recommended values include: 1) Material differences, such as polyester, cotton, and spandex, whose composition and proportion significantly affect the warmth of the clothing, with significant differences in insulation values between jackets of different materials; 2) Thickness differences, the standards do not clearly define “thin” and “thick”, leading to possible differences in insulation values even for the same type of clothing; 3) Style differences, such as hooded and non-hooded sweatshirts, long and short down coats, and other style variations also affect insulation values. In addition, the thermal resistance recommended values in the ISO 7730-2005 and ASHRAE 55-2020 standards are mainly based on typical Western clothing from the 20th century and may not be applicable to modern diverse types of clothing.

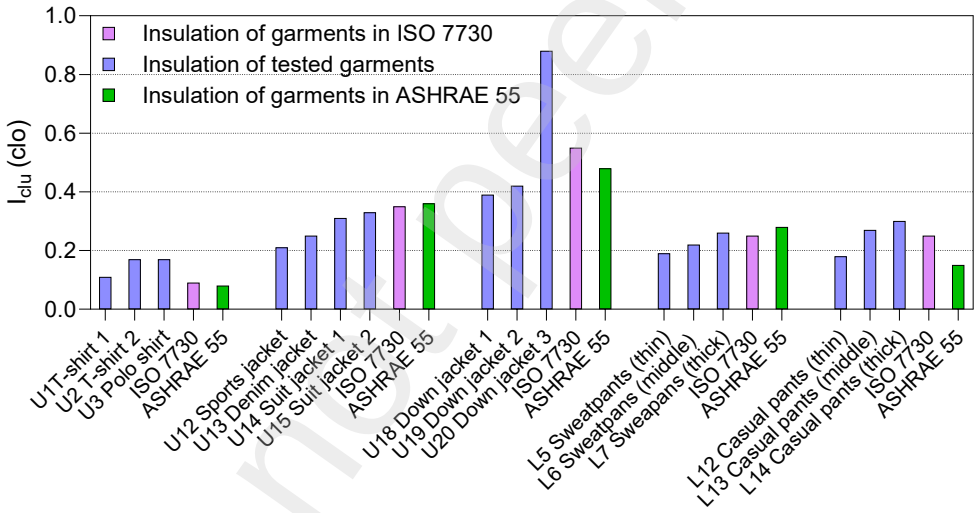


Fig. 12. Comparison of the clothing insulation between tested results and standards

4.2. Comparing the estimation methods of ensembles clothing insulation

Building on existing standards [16], the most common method to obtain clothing insulation values in thermal comfort field research is to sum individual garments and calculate the overall suit insulation using a correction equation. The ISO 9920-2007 [16] standard provides a calculation method developed by McCullough et al. [45], with the formula (Eq. 6) applicable to a range of 0.2 to 1.6 clo and widely used in field studies [49,50].

Tang et al. [38] based on actual measurements of single and suit clothing insulation for the elderly population in China, provided a linear regression equation for the overall

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77suit versus the sum of individual garments, as shown in Eq. 7, mainly targeting garments above 1.0 clo.

$$I_{cl} = 0.972 \sum I_{clu,i} + 0.168, R^2 = 0.912 \quad (7)$$

$\sum I_{clu,i}$ represents the sum of individual garment insulations in clo.

To analyze the applicability of traditional methods, Fig. 13 compares the measured results of suit clothing insulation in this study with the predicted values calculated by the McCullough et al. [45] and Tang et al. [38] models, and Table 4 provides statistical results and error analysis. Data analysis shows that there is a significant difference between the predicted values of the McCullough et al. equation and the measured values, with average prediction errors of 19%, 10%, and 11% for sports ensembles, office ensembles, and casual ensembles, respectively. Overall, the prediction error of the McCullough et al. model in this study increases with the increase in clothing insulation. When the ensemble clothing insulation is below 0.6 clo, the measured values are lower than the predicted values; when the insulation value is greater than 1.0 clo, the predicted values are generally lower than the measured values, with a maximum error of 0.28 clo. The standard indicates that the McCullough et al. equation is applicable for a range of 0.2 to 1.6 clo, but the results show that it is not suitable for calculating the insulation values of adult Chinese ensembles. In summer, the predicted values overestimate the measured values, while in winter, they underestimate the measured values.

The Tang et al. [38] model's test data mainly focuses on clothing insulation above 1.0 clo. Comparisons between the model and measured values reveal that as clothing insulation increases, the error rate gradually decreases. When clothing insulation is below 1.0 clo, the average absolute error of the Tang et al. model is 15%, and when insulation is above 1.0 clo, the average absolute error drops to 5.5%. This indicates that for high clothing insulation values, the prediction accuracy of the Tang et al. model is significantly higher than that of the McCullough et al. [45] model.

A substantial amount of thermal comfort field research has shown a "scissor gap" phenomenon between PMV and Thermal sensation vote (TSV) [46,47]: in warm environments, PMV predicted values are higher than actual thermal sensations; in cool environments, PMV predicted values are lower than actual thermal sensations. This suggests that errors in clothing insulation input values contribute to the "scissor gap" phenomenon. Our measured results also confirm this: in warm environments, overestimated clothing insulation leads to PMV predicted values higher than actual thermal sensations; in cool environments, underestimated clothing insulation leads to PMV predicted values lower than actual thermal sensations.

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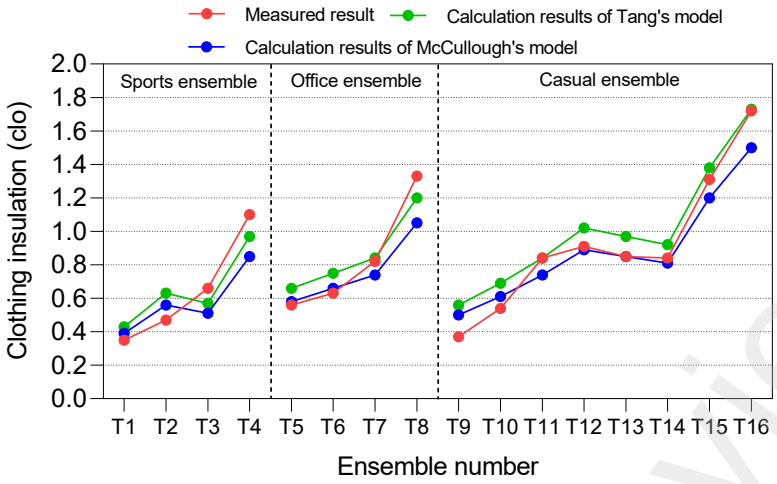


Fig. 13. Comparison of the measured results and the predicted results of McCullough et al. [45] and Tang et al. [38] model

Table 4 Statistical results and error analysis of McCullough et al. and Tang et al. model.

Number	Measured value	McCullough et al. model		Tang et al. model	
		Predicted value	Error[#]	Predicted value	Error
T1	0.35	0.39	11%	0.43	24%
T2	0.47	0.56	19%	0.63	3%
T3	0.66	0.51	23%	0.57	7%
T4	1.10	0.85	23%	0.97	9%
T5	0.56	0.58	4%	0.66	22%
T6	0.63	0.66	5%	0.75	11%
T7	0.82	0.74	10%	0.84	4%
T8	1.33	1.05	21%	1.20	10%
T9	0.37	0.5	35%	0.56	36%
T10	0.54	0.61	13%	0.69	24%
T11	0.84	0.74	12%	0.84	15%
T12	0.91	0.89	2%	1.02	10%
T13	0.85	0.85	0%	0.97	11%
T14	0.84	0.81	4%	0.92	12%
T15	1.31	1.2	8%	1.38	3%
T16	1.72	1.5	13%	1.73	0%

([#]=|measure value - estimated average value|/measured value)

Existing models do not consider the diversity of clothing types, which may lead to significant deviations in the estimated ensemble clothing insulation. To address this, our study has established linear regression equations for the ensemble clothing insulation of different types based on actual measurement data, as shown in Fig. 14. Compared to the McCullough et al. [45] model and the Tang et al. [38] model, the predictive equations we proposed significantly reduced the prediction bias, with an

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average prediction bias controlled within 10%. Tang et al. [38] research indicates that although the overall linear prediction equation has improved the prediction to some extent, the prediction error remains large in cases of high clothing insulation, suggesting that the overall linear prediction equation may not be suitable for predicting high ensemble clothing insulation, mainly due to the uneven distribution of local clothing insulation. Therefore, future research should increase the sample size of different types of clothing and establish more accurate local prediction equations to estimate the ensemble clothing insulation [38,39,51], thereby improving the accuracy of predicting the insulation of suit clothing from individual garments.

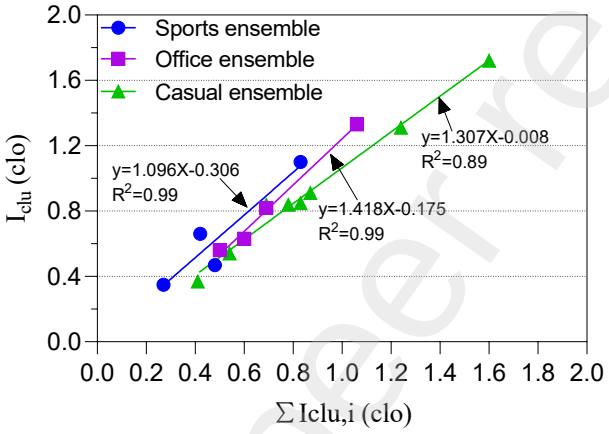


Fig. 14. Liner regression of ensembles' overall insulation against the sum of garments' overall insulation.

4.3. Influence of clothing insulation estimation error on PMV prediction value

Due to its flexibility and convenience, the PMV model has become a widely used model for evaluating indoor thermal environments and human thermal comfort [20,21,52]. The clothing insulation value is a key input parameter in the PMV model. To quantify the impact of estimated and measured clothing insulation on the PMV model's calculation results, we input various sets of estimated and measured clothing insulation values into the PMV model calculation program for comparative analysis [12]. In actual surveys, we found a significant correlation between clothing insulation and air temperature [31,32]. Drawing on past research, we made the following assumptions about the variation of clothing insulation and the corresponding air temperature settings: when the ensemble clothing insulation is less than 0.7 clo, the air temperature is set to 26 °C (summer); when the insulation is between 0.7 and 1.0 clo, the air temperature is set to 22 °C (transition season); when the insulation is greater than 1.0 clo, the air temperature is set to 18 °C (winter). Under these settings, the air velocity is 0.1 m/s, the metabolic rate is 1.1 met (representing a sedentary office state), the average radiant temperature is the same as the air temperature, and the relative humidity is maintained

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at 50%. Under the same temperature conditions, all other input variables remain the same except for clothing insulation to assess the impact of estimation errors in clothing insulation on the accuracy of the PMV prediction model.

Fig. 15 illustrates the impact of estimated and measured ensemble clothing insulation on PMV calculation results. Table 5 provides statistical results for PMV values calculated based on estimated and measured clothing insulation. The calculation results show that in summer conditions where clothing insulation is less than 0.7 clo, the error in PMV caused by different researchers' estimations of clothing insulation is within 0.5 units. As air temperature decreases and clothing insulation increases (transition season and winter conditions), the error in PMV calculation results due to estimation results can reach up to 1.3 units, and the maximum difference between PMV values calculated from estimated clothing insulation and measured values is 0.59 units. This result emphasizes the importance of inputting accurate clothing insulation values in PMV calculations and shows that the lower the environmental temperature, the more significant the impact of clothing insulation on thermal comfort. This could lead to incorrect building operation strategies, resulting in more discomfort for the subjects and energy waste.

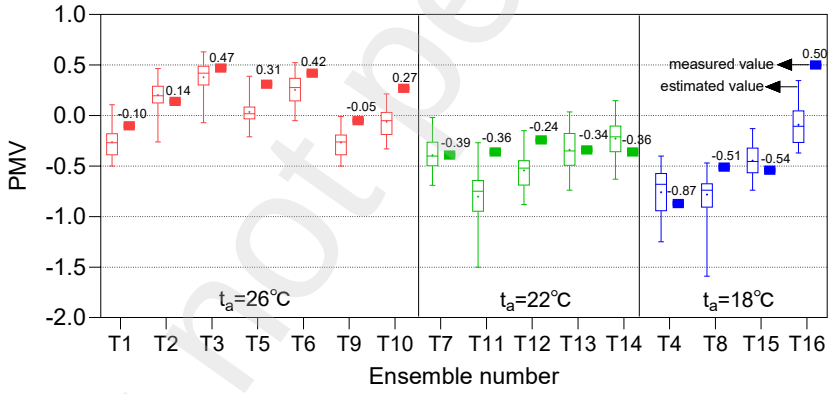


Fig. 15. Influences of clothing insulation estimation and measured value error on PMV model (the box plot shows the PMV value calculated from the insulation values estimated by different researchers, the square represents the PMV value calculated from the measured insulation values)

Table 5 Statistical table of PMV values calculated from estimation and measured clothing insulation value.

Number	Estimated value			Measured value	Error value[#]
	Minimum	Maximum	Average ± SD		
T1	-0.5	0.11	-0.26±0.14	-0.10	0.16
T2	-0.26	0.47	0.20±0.15	0.14	0.06
T3	-0.07	0.63	0.38±0.15	0.47	0.09
T5	-1.25	-0.40	-0.76±0.23	0.31	1.07

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T6	-0.21	0.389	0.04±0.13	0.42	0.38
T9	-0.05	0.52	0.25±0.14	-0.05	0.30
T10	-0.69	-0.02	-0.39±0.17	0.27	0.66
T7	-1.59	-0.47	-0.78±0.20	-0.39	0.39
T11	-0.5	-0.01	-0.27±0.12	-0.36	0.09
T12	-0.33	0.22	-0.06±0.15	-0.24	0.18
T13	-1.5	-0.27	-0.80±0.25	-0.34	0.46
T14	-0.88	-0.15	-0.54±0.17	-0.36	0.18
T4	-0.74	0.04	-0.34±0.21	-0.87	0.53
T8	-0.63	0.15	-0.23±0.19	-0.51	0.28
T15	-0.74	-0.13	-0.45±0.15	-0.54	0.09
T16	-0.37	0.35	-0.09±0.18	0.50	0.59

(# is estimated average value minus measured value)

4.4. Influence of clothing insulation estimation error on T_{sk} and SET

The Standard Effective Temperature (SET) model is a pivotal evaluation criterion in thermal comfort research and building environmental design [16,53], integrating the impact of human thermoregulation and environmental parameters. When applying the SET model, clothing insulation as a key input parameter significantly influences the model's computational outcomes. To quantify the impact of estimated versus measured clothing insulation on the SET model results, we conducted a comparative analysis by inputting various sets of estimated and measured clothing insulation values into the SET model.

Referring to the analysis method of the PMV model, we made assumptions about the variation of clothing insulation and the corresponding air temperature settings: when clothing insulation is less than 0.7 clo, the air temperature is set to 26 °C, representing summer conditions; when clothing insulation is between 0.7 and 1.0 clo, the air temperature is set to 22 °C, representing transitional season conditions; when clothing insulation exceeds 1.0 clo, the air temperature is set to 18 °C, representing winter conditions. Under these settings, the air velocity is maintained at 0.1 m/s, the metabolic rate is set to 1.1 met, representing a sedentary office state, the average radiant temperature is the same as the air temperature, and the relative humidity is maintained at 50%. With all other input variables held constant, we assessed the impact of estimation errors in clothing insulation on the accuracy of the SET prediction model.

As shown in Fig. 16, variations in the estimation of ensemble clothing insulation significantly affect the SET calculation results. In the summer conditions where clothing insulation is less than 0.7 clo, the estimation of clothing insulation by different researchers results in SET errors within 2 °C. In approximately 75% of cases, the SET

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results calculated from the average estimated clothing insulation are lower than those calculated from the measured values. Furthermore, the larger the clothing insulation value, the greater the error in SET values calculated by different researchers. In winter conditions, the error in SET calculation results can reach up to 5 °C.

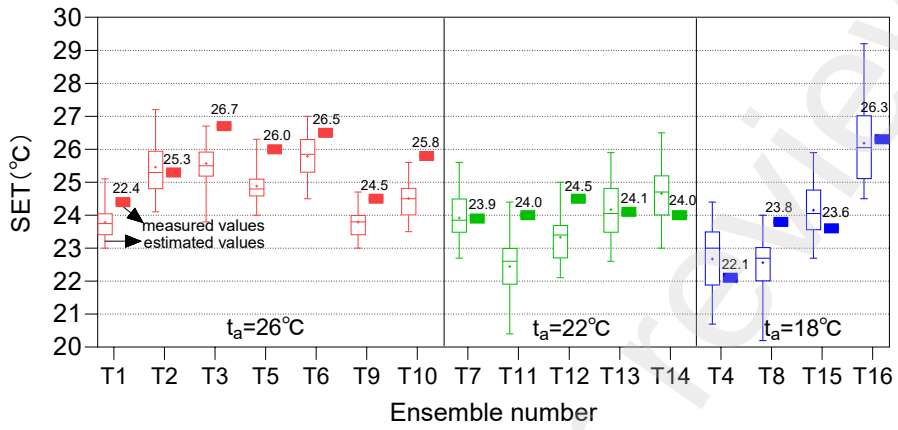


Fig. 16. Influences of clothing insulation estimation and measured value error on SET model

In regulating human thermal comfort, the interaction between clothing insulation and skin temperature plays an essential role [54]. Variations in clothing insulation can significantly affect skin temperature, influencing human thermal comfort. We used the SET model to simulate the skin temperatures corresponding to estimated and measured clothing insulation at different temperatures, as shown in Fig. 17. Table 6 provides statistical data on skin temperatures calculated based on estimated and measured clothing insulation. The analysis results show that for the same piece of clothing, the greater the standard deviation of thermal resistance estimated by different researchers, the greater the variability in skin temperature predicted by the SET model. At an ambient temperature of 26 °C, the variability between estimated and measured skin temperatures is relatively small, with the standard deviation of skin temperature due to different researchers' estimates kept within 0.1 °C. The largest difference occurs when the clothing insulation is greater than 1.0 clo, where the skin temperature errors due to different researchers' insulation estimates can reach 1.74 °C. As the air temperature decreases and the ensemble clothing insulation increases, the difference between estimated and measured skin temperatures significantly increases, with the maximum error at the suit T11 location being 0.68 °C, and all statistically showing significant differences. The comparative results indicate that accurate input of clothing insulation is significant for predicting human thermal responses and heat loads.

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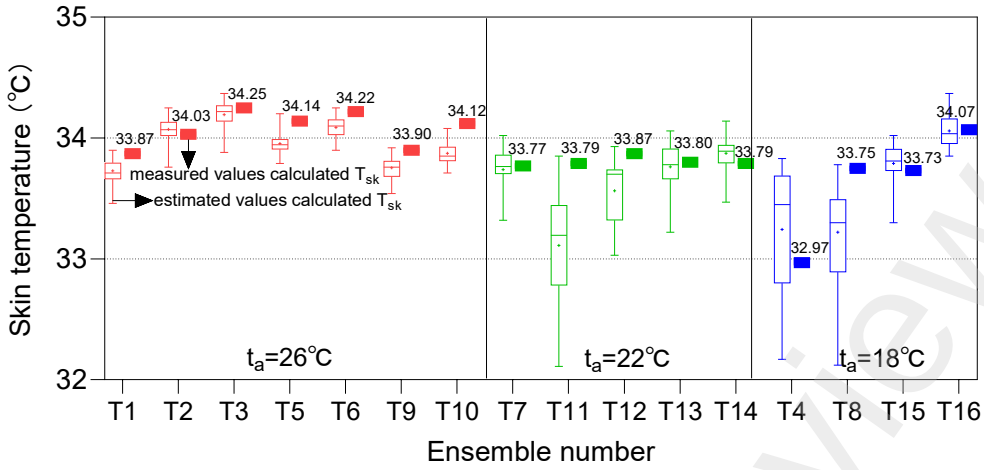


Fig. 17. Influences of clothing insulation estimation and measured value error on skin temperature

Table 6 Statistical table of skin temperature calculated from estimation and measured clothing insulation value.

Number	Estimated value (℃)			Measured value (℃)	Error value[#] (℃)
	Min	Max	Average ± SD		
T1	33.46	33.9	33.73±0.09	33.87	-0.14
T2	33.76	34.25	34.07± 0.10	34.03	0.04
T3	33.88	34.37	34.19±0.10	34.25	-0.06
T5	33.79	34.2	33.96±0.09	34.14	-0.18
T6	33.90	34.25	34.09±0.09	34.22	-0.13
T9	33.54	33.92	33.76±0.08	33.9	-0.14
T10	33.71	34.08	33.87±0.09	34.12	-0.25
T7	33.32	34.02	33.74±0.17	33.77	-0.03
T11	32.11	33.85	33.11±0.46	33.79	-0.68
T12	33.03	33.93	33.56±0.24	33.87	-0.31
T13	33.22	34.06	33.76±0.20	33.8	-0.04
T14	33.47	34.14	33.87±0.14	33.79	0.08
T4	32.17	33.83	33.24±0.51	32.97	0.27
T8	32.12	33.78	33.22±0.40	33.75	-0.53
T15	33.30	34.02	33.79±0.14	33.73	0.06
T16	33.85	34.37	34.06±0.13	34.07	-0.01

([#] is the estimated average value minus the measured value.)

4.5. limitation and further research

Accurate clothing insulation data is crucial for designing and regulating building environmental control systems, aiding in more effective energy utilization and optimization of indoor thermal environments. This study compared the estimated clothing insulation obtained by different scholars using the lookup table method with actual measured values. It explored the potential impact of this difference on model

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prediction results, but the study has some limitations. Firstly, we collected 120 survey questionnaires from different research institutions and measured 34 garments and 16 ensembles, quantifying the difference between estimated and measured values to some extent. However, we recognize that the types of clothing and sample sizes measured are limited, which may affect the universal applicability of the results.

The development of infrared thermography technology can directly measure the distribution of temperature fields of clothing in different environments while worn. Images can be collected for temperature difference analysis and data processing to monitor the insulation of indoor personnel's clothing in real-time. This technology is more precise than the lookup table method and can be used in future studies to measure indoor personnel's clothing insulation dynamically. Secondly, in thermal comfort field surveys, we observed that subjects adjust their clothing behavior according to the environment, such as unzipping zippers or rolling up sleeves. These behaviors may change the actual clothing insulation value, thereby affecting the perception of thermal comfort and the model's prediction accuracy. Therefore, future studies need to consider these dynamic clothing behavior factors. By testing the clothing insulation values in different dressing states and comparing them with the estimates of clothing insulation by different researchers, we can further quantify the differences between estimated and measured values. This will provide a scientific basis for the revision of relevant thermal comfort and clothing standards and provide empirical data and directions for improvement to enhance the prediction accuracy of thermal comfort models.

5. Conclusion

This study selected 20 upper-body and 14 lower-body garments, combining them into 16 types of clothing sets. Through questionnaires, we collected the clothing insulation estimates from 120 researchers based on standard calculations. Concurrently, we used a thermal manikin in a climatic chamber to precisely measure the insulation values of individual garments and ensembles. The following conclusions were drawn:

(1) A comprehensive clothing insulation dataset was established for 34 single garments and 16 ensembles (including office, sports, and casual ensembles), providing both overall and local insulation values. This dataset enhances the representativeness of current thermal insulation standards.

(2) Significant estimation errors were found across participants, particularly for garments with higher insulation. The maximum deviation between estimated and measured values reached 69% for upper-body items and 30% for lower-body garments. In contrast, the maximum error for sets reached 0.21 clo, with an error rate of 33%.

(3) Substantial differences were identified between measured insulation values and

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those recommended in ISO 7730 and ASHRAE 55. These discrepancies are influenced by variations in fabric material, thickness, and garment style. Moreover, the commonly used McCullough equation was found unsuitable for Chinese adult ensembles, leading to overestimations in summer and underestimations in winter.

(4) When input into the PMV model, estimated insulation values produced prediction errors up to 1.3 units, with a maximum deviation of 0.59 units compared to predictions using measured values. These discrepancies can affect thermal comfort classification and HVAC system decisions.

(5) Comparing the impact of estimated and measured clothing insulation on SET values and human skin temperature, it was discovered that the estimates from different researchers led to SET model calculation errors ranging between 2 to 5°C, and inaccurate estimation of insulation values could result in an average skin temperature deviation of up to 1.74 °C.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A: Characteristics and clothing insulation of measured garments

Code	Garment description	Component	Weight (kg)	I_{cl} (clo)	I_{clu} (clo)
U1	T-shirt 1	100%polyester fiber	0.122	0.140	0.110
U2	T-shirt 2	100%cotton	0.220	0.200	0.170
U3	Polo shirt	100%cotton	0.250	0.220	0.170
U4	Short-sleeved shirt	100%polyester fiber	0.200	0.220	0.180
U5	Sports long sleeve 1	100%polyester fiber	0.236	0.140	0.120
U6	Sports long sleeve 2	100%polyester fiber	0.248	0.160	0.130
U7	Long-sleeved underwear	100%cotton	0.262	0.220	0.180
U8	Long-sleeved shirt 1	100%cotton	0.268	0.280	0.240
U9	Long-sleeved shirt 2	100%cotton	0.268	0.240	0.240
U10	Sweatshirt	80%cotton+20%polyester fiber	0.384	0.290	0.240
U11	Hoodies	82%cotton+18%polyester fiber	0.568	0.265	0.220
U12	Sports jacket	100%nylon	0.388	0.25	0.210
U13	Denim jacket	93%cotton+7%polyester fiber	0.830	0.300	0.250
U14	Suit jacket 1	100%polyester fiber	0.620	0.370	0.310
U15	Suit jacket 2	100%polyester fiber	0.580	0.390	0.330
U16	Sweater	66%cotton+34%polyester fiber	0.610	0.340	0.280
U17	Outdoor jacket	100%polyester fiber	0.930	0.450	0.380
U18	Down jacket 1	100%polyester fiber+75%down feather	0.634	0.460	0.390
U19	Down jacket 2	100%polyamide+90%down feather	0.380	0.490	0.420
U20	Down jacket 3	100%polyester fiber+80%down feather	1.106	1.030	0.880
L1	Shorts	100%polyester fiber	0.118	0.080	0.070
L2	Light skirt	85%polyester fiber+15%spandex	0.160	0.250	0.210
L3	Long legs underwear	100%cotton	0.242	0.270	0.220
L4	Sweatpants + shorts	100%polyester fiber	0.216	0.170	0.150
L5	Sweatpants (thin)	80%nylon+20%spandex	0.376	0.220	0.190
L6	Sweatpants (middle)	100%cotton	0.488	0.320	0.260
L7	Sweatpants (thick)	100%cotton	0.448	0.260	0.220
L8	Coveralls	98%cotton+2%spandex	0.480	0.290	0.240
L9	Suit pants	100%terylene	0.380	0.280	0.250
L10	Jeans (thin)	93%cotton+7%terylene	0.564	0.260	0.210
L11	Jeans (thick)	93%cotton+5%polyester fiber+2%terylene	0.450	0.290	0.240
L12	Casual pants (thin)	100%cotton	0.298	0.230	0.180
L13	Casual pants (middle)	100%polyester fiber	0.250	0.320	0.270
L14	Casual pants (thick)	100%cotton	0.545	0.350	0.300

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Appendix B: Overall and local insulation values test results of clothing ensembles

Code	Ensembles	Whole body	Upper arm	Fore arm	Hand	Chest	Shoulders	Stomach	Back	Hip front	Hip back	Thigh front	Thigh back	Calf front	Calf back	Foot
T1	U1+ L1	0.95	0.36	0.03	0.00	0.45	0.62	1.46	1.15	2.89	2.34	0.53	0.47	0.00	0.00	0.64
T2	U5+L4	1.05	0.32	0.35	0.10	0.58	0.64	1.44	1.65	1.82	2.20	0.51	0.57	0.19	0.15	0.68
T3	U6+L5	1.23	0.62	0.57	0.16	0.69	0.79	1.53	1.25	2.26	2.21	1.21	1.38	0.47	0.36	0.61
T4	U6+U17+L6	1.60	1.97	2.23	0.06	2.20	3.08	4.52	3.46	3.76	3.66	1.00	1.12	0.77	0.58	0.66
T5	U3+L12	1.13	0.55	0.05	0.02	0.52	0.51	1.04	1.14	2.14	3.07	0.85	1.07	0.75	0.52	0.54
T6	U8+L12	1.14	0.69	0.73	0.03	0.60	0.75	1.27	1.10	1.94	2.52	0.63	0.61	0.57	0.39	0.56
T7	U8+U15+L9	1.36	1.17	1.24	0.07	1.21	1.63	2.58	2.11	3.34	3.12	0.57	0.88	0.84	0.55	0.56
T8	U19+U8+L14	1.81	3.00	3.71	0.20	2.90	3.95	5.63	4.36	5.05	5.71	0.92	1.38	0.96	0.81	0.72
T9	U2+L1	0.97	0.36	0.04	0.03	0.34	0.36	0.97	0.93	2.15	2.37	0.84	0.81	0.02	0.02	0.73
T10	U2+L2	1.11	0.51	0.06	0.04	0.79	0.44	1.31	1.49	2.73	3.70	1.46	1.46	0.13	0.09	0.62
T11	U9+L10	1.37	0.98	0.92	0.16	1.07	1.23	2.36	2.10	3.11	3.24	1.06	0.98	0.56	0.40	0.77
T12	U4+U14+L13	1.44	1.05	0.98	0.07	1.00	1.25	2.54	2.24	4.32	3.66	1.36	1.79	0.65	0.57	0.86
T13	U10+U12+L7	1.38	1.50	1.51	0.12	1.32	1.65	3.13	1.99	2.25	1.91	1.15	1.41	0.51	0.35	0.69
T14	U11+U13+L8	1.37	0.95	1.09	0.16	1.12	1.55	1.98	1.80	2.50	2.12	0.85	1.60	0.69	0.53	0.78
T15	U18+U10+L3+L6	1.79	2.95	3.23	0.15	3.03	4.66	4.86	4.12	3.02	4.17	1.18	0.85	1.07	0.76	0.83
T16	U7+U16+U20+L3 +L11	2.17	3.42	4.03	0.19	3.61	5.46	6.43	4.83	6.18	6.73	3.49	3.26	1.23	0.82	0.86

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